A

Report of Mini Project

On

"IoT-Based Low-Cost Multi-Vital Sign Monitoring Device for Rural Telemedicine"

Submitting In Partial Fulfillment of the Requirement for the Bachelor of Degree in Third Year
Electronics and Telecommunication Engineering
Dr. Babasaheb Ambedkar Technological University, Lonere

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CERTIFICATE

This is to certify that the Report of Mini Project entitled

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For Academic Year 2025-26

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1. INTRODUCTION

In recent years, the rapid advancement of the Internet of Things (IoT) has revolutionized various sectors, including healthcare. The integration of IoT into medical systems has paved the way for smart health monitoring and telemedicine solutions, which enable doctors to monitor patients' health conditions remotely and provide timely medical assistance. With the rise of chronic diseases and the increasing demand for continuous healthcare, IoT-based monitoring has emerged as a powerful tool to ensure efficient, real-time, and patient-centric medical care.

In rural and underdeveloped regions, access to healthcare facilities and professional medical support remains a major challenge. Patients often travel long distances to hospitals for basic health check-ups, leading to delayed diagnosis and treatment. To overcome these limitations, remote health monitoring systems are being developed to bridge the gap between patients and healthcare providers.

The Internet of Things offers a reliable platform for connecting medical devices and sensors to the internet, allowing vital health data such as heart rate, SpO_2, ECG, and body temperature to be measured, analyzed, and transmitted to healthcare professionals. This not only reduces the dependency on hospital visits but also facilitates continuous observation of patient health conditions, even from remote locations.

This project focuses on designing and developing a low-cost, portable, and battery-powered IoT-based health monitoring system capable of acquiring real-time physiological data using multiple biomedical sensors. The collected data is processed by the ESP8266 microcontroller and transmitted to the ThingSpeak IoT cloud platform, where it can be visualized and analyzed by doctors or caretakers.

By leveraging wireless communication, sensor technology, and cloud computing, this system enables real-time data sharing, continuous patient monitoring, and effective telemedicine support, particularly for rural healthcare centers and home-based patients. The proposed system demonstrates how IoT can transform conventional healthcare into an accessible, connected, and data-driven ecosystem, ultimately improving the quality of life for patients.

2. LITERATURE SURVEY

S. Islam et al., “The Internet of Things for Health Care: A Comprehensive Survey,” IEEE Access, 2015.This paper provided a broad understanding of IoT applications in healthcare, including system architectures, communication technologies, and data management strategies. From this survey, the foundational concept of integrating IoT with biomedical sensors for remote health monitoring was adopted, helping to structure the overall system design and data flow in the project.

S. K. Sood et al., “IoT-based smart healthcare kit for diabetic patients,” Future Generation Computer Systems, 2018.This research inspired the use of a modular, low-cost, and sensor-based health monitoring kit. The approach of continuous physiological data acquisition and real-time cloud storage was adapted for monitoring multiple vital signs such as heart rate, SpO_2, and temperature in the developed system.

A. K. Tyagi et al., “IoT-based health monitoring system for remote and rural areas,” IJERT, 2019.This paper emphasized the importance of developing affordable IoT-based health systems for rural and remote regions. It influenced the project’s goal to create a low-cost, portable solution that can bridge the healthcare accessibility gap for patients in underserved areas..

R. J. Silva et al., “A low-cost portable health monitoring system using IoT and cloud computing,” IEEE Latin America Transactions, 2021.This study provided insight into combining IoT with cloud computing for efficient health data transmission and storage. The project utilized similar cloud-based techniques for uploading patient data to the ThingSpeak platform, ensuring real-time access and remote monitoring.

P. Mehta et al., “Edge-based health monitoring system using ESP32 and ThingSpeak for rural areas,” Procedia Computer Science, 2023.This paper directly influenced the hardware and platform selection in the project. The use of ESP32 and ThingSpeak for data transmission and visualization was adapted to enhance system efficiency, connectivity, and real-time health tracking capabilities.

3. OBJECTIVE

The objectives of the project can be summarized as follows:

1. To develop a compact, affordable, and energy-efficient hardware system using microcontrollers and biomedical sensors.
2. To integrate multiple sensors for simultaneous monitoring of vital signs such as ECG, SpO_2, heart rate, and body temperature.
3. To implement IoT connectivity using the ESP8266 Wi-Fi module for real-time transmission of health data to cloud storage.
4. To design a cloud-based monitoring interface on ThingSpeak for visualizing and analyzing health parameters.
5. To provide early alerts and continuous observation of patient vitals, reducing the need for frequent hospital visits.
6. To support telemedicine and remote healthcare services, improving the quality of healthcare delivery in resource-constrained regions.
7. To ensure user-friendly operation, portability, and reliability, making it suitable for home-based and field healthcare applications.

Overall, this project seeks to bridge the gap between patients and healthcare professionals by providing a smart, connected, and cost-effective medical monitoring solution. The system not only contributes to early disease detection and preventive healthcare but also demonstrates the potential of IoT in revolutionizing the field of rural telemedicine and e-healthcare.

4. METHODOLOGY

The proposed system is based on the integration of IoT technology with biomedical sensors to design a compact, low-cost, and portable multi-vital sign monitoring device. The methodology focuses on the hardware design, data acquisition, data transmission, and cloud-based visualization processes to ensure accurate and real-time monitoring of a patient's health status.

The system uses the ESP8266 NodeMCU as the core microcontroller unit (MCU), which serves as both the data processor and the communication interface. Multiple sensors are interfaced with the NodeMCU to acquire different vital parameters:

AD8232 ECG Sensor – captures the electrical activity of the heart to monitor ECG waveforms.

MAX30100 Sensor – measures heart rate and SpO_2 (oxygen saturation) levels using photoplethysmography.

LM35 Temperature Sensor – detects body temperature with high accuracy.

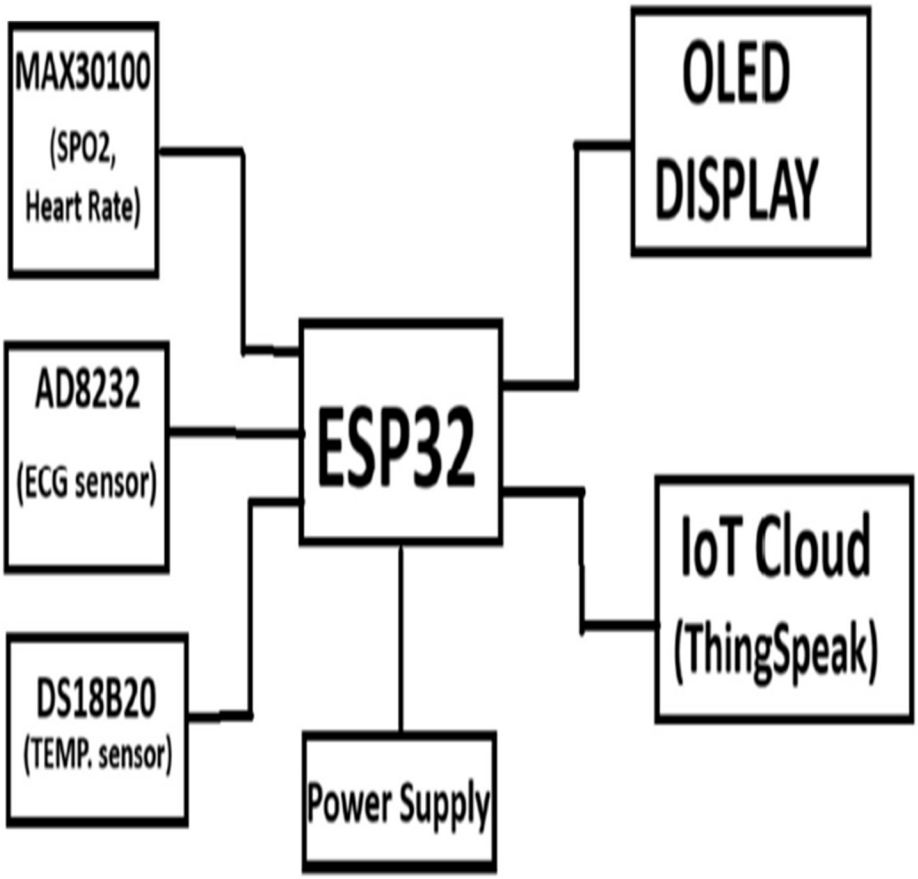
The process begins with the sensors continuously acquiring physiological signals from the patient. These analog signals are converted into digital form and processed by the ESP8266. The microcontroller then formats the data and uploads it to the cloud platform (ThingSpeak) using its built-in Wi-Fi module.

On the ThingSpeak IoT cloud, the data is stored, analyzed, and displayed graphically in real time. Doctors or healthcare professionals can remotely access the cloud dashboard from anywhere using an internet-enabled device to observe patient health trends and detect abnormalities promptly.

To make the device portable and user-friendly, it is powered by a rechargeable lithium-ion battery with a regulated power management system to ensure stable operation. An OLED display module is also incorporated to provide instant local feedback of sensor readings to the patient.

The overall methodology ensures that the system operates efficiently in remote locations with limited infrastructure. By combining wireless communication, cloud computing, and sensor technology, the proposed model delivers a cost-effective and reliable health monitoring solution suitable for rural telemedicine applications.

4.1 BLOCK DIAGRAM



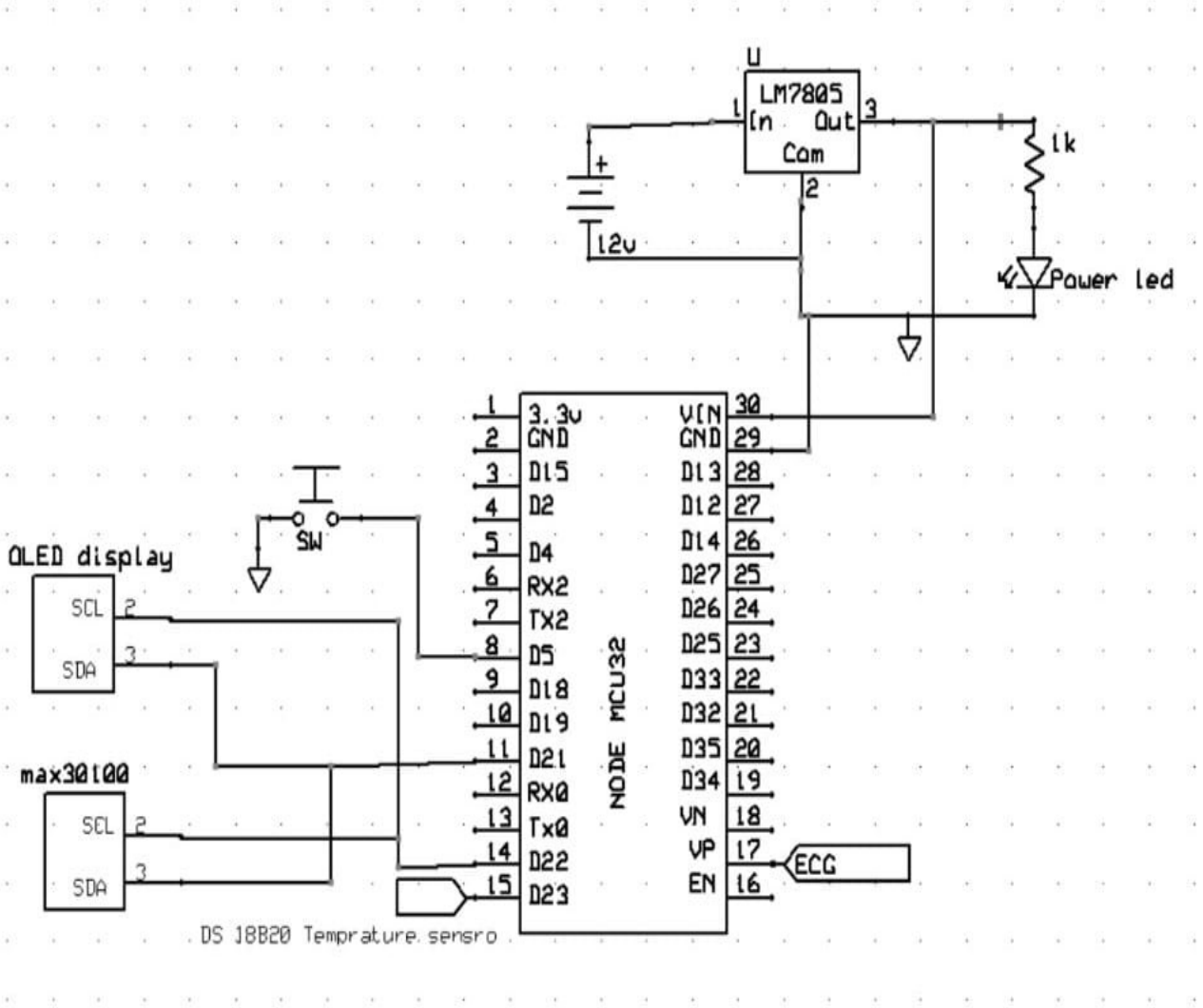
➢ Block Diagram Description

The block diagram illustrates the working of the IoT-based multi-vital sign monitoring system, which consists of three main sections: data acquisition, processing & display, and remote monitoring.

In the data acquisition stage, biomedical sensors such as the MAX30100 (Heart Rate & SpO_2), AD8232 (ECG), and DS18B20 (Temperature) collect vital physiological parameters. The collected data is then sent to the ESP32 microcontroller, which processes the signals and displays the results on an OLED display for real-time monitoring.

For remote monitoring, the processed data is uploaded to the ThingSpeak IoT cloud platform, allowing doctors to access and analyze patient health information from anywhere. Additionally, IFTTT integration enables automated email/SMS alerts in case of abnormal readings, ensuring timely medical response.

4.2 CIRCUIT DIAGRAM



4.3 WORKING PRINCIPLE

An IoT-Based Low-Cost Multi-Vital Sign Monitoring Device for Rural Telemedicine works on the principle of continuous sensing, digital processing, wireless transmission, and remote monitoring of multiple physiological parameters of a patient. Biomedical sensors such as ECG, pulse oximeter (SpO_2), temperature sensor, and accelerometer are attached to the patient's body to measure vital signs including heart rate, ECG waveform, oxygen saturation level, body temperature, and physical movement. These sensors detect physical or biological changes in the body and convert them into weak analog electrical signals.

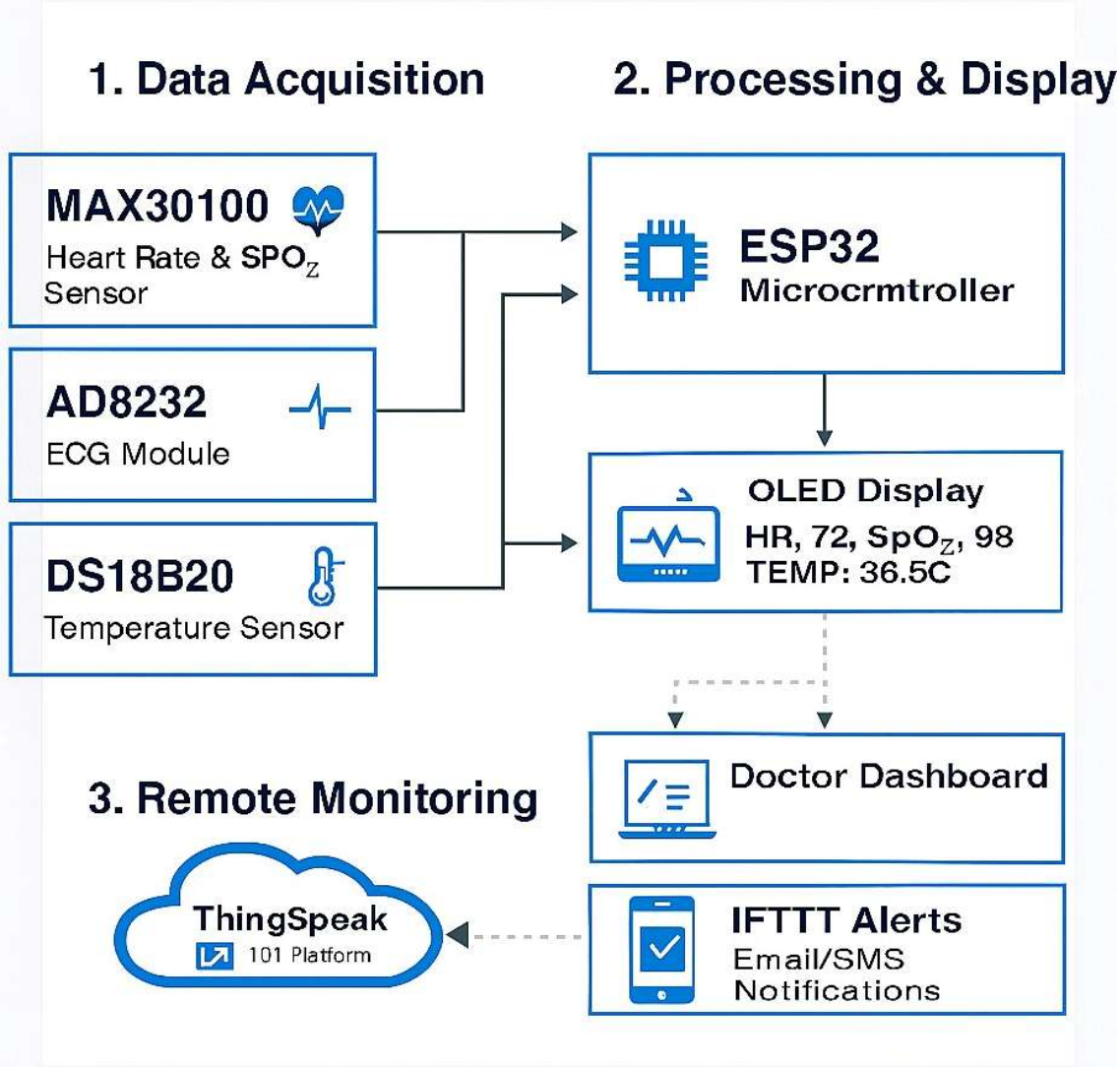
These raw analog signals are usually very small and affected by noise and disturbances. Therefore, they are passed through signal conditioning circuits, which include amplifiers, filters, and noise reduction units to enhance signal strength and remove unwanted noise. After conditioning, the analog signals are fed into the Analog-to-Digital Converter (ADC) present in the microcontroller (such as ESP32 or Arduino), where they are converted into digital signals.

The microcontroller then performs data processing, feature extraction, and threshold comparison to identify important physiological parameters such as heart rate (from ECG), oxygen level and pulse rate (from SpO_2 sensor), and body temperature readings from the temperature sensor. If any parameter goes beyond normal preset limits, the microcontroller detects this as an abnormal or emergency condition.

The processed data is displayed locally on a LCD or OLED screen, and simultaneously it is transmitted wirelessly to a cloud server using IoT communication technologies like Wi-Fi, GSM, or MQTT protocols. From the cloud, the data can be accessed by doctors, caregivers, or healthcare professionals through a mobile app or web dashboard from any remote location.

If abnormal readings are detected, the system automatically sends alerts, SMS, or notifications to the doctor or caretaker, enabling timely medical attention. Thus, the system enables real-time, low-cost, and remote health monitoring, especially suitable for rural areas where access to hospitals and medical professionals is limited.

4.4 FLOW CHART



➢ Flow Chart Description

The circuit diagram represents the hardware interconnection of the system components. The ESP32 microcontroller serves as the central processing unit, interfacing with the MAX30100, DS18B20, AD8232 ECG sensor, and OLED display. Communication between the ESP32 and the sensors/display occurs via I²C and digital/analog interfaces. A LM7805 voltage regulator converts the 12V supply to a stable 5V output to power the system, while a power LED indicates circuit activity.

In operation, the sensors acquire vital data, which is processed by the ESP32 and displayed locally. Simultaneously, the same data is transmitted wirelessly to the cloud for continuous remote monitoring and alert generation.

4.5 HARDWARE INPUTS

Component	Function	Purpose in System
ESP32 Microcontroller	Main Processing & Wi-Fi	Controls sensors, processes data, and uploads to IoT cloud
MAX30100 Sensor	SpO_2 & Heart Rate Sensor	Measures oxygen saturation and pulse rate
AD8232 ECG Sensor	ECG Signal Acquisition	Captures electrical activity of the heart
DS18B20 Temperature Sensor	Digital Temperature Sensor	Measures body temperature accurately
OLED Display	Output Display	Shows real-time health parameters
Power Supply (Battery)	Power Source	Provides operating voltage to ESP32 and sensors

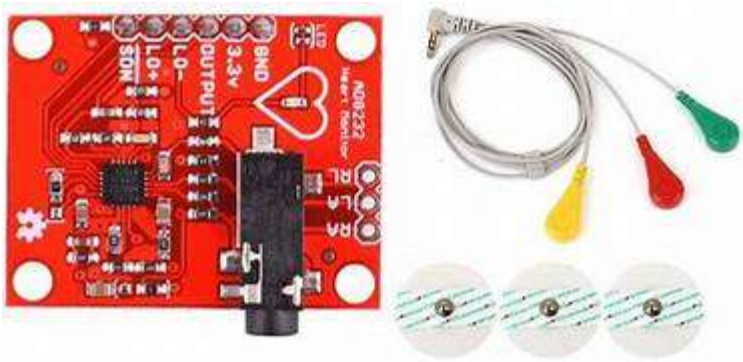
1. **ESP32 Microcontroller :**
   ESP32 is a low-cost, low-power, and high-performance microcontroller developed by Espressif Systems. It is widely used in IoT applications, wearable devices, health-monitoring systems, smart home automation, and industrial solutions due to its integrated Wi-Fi and Bluetooth connectivity. The ESP32 is highly rated for its versatility, reliability, and strong community support, making it one of the most preferred microcontrollers for modern IoT development.



2. **MAX30100 Sensor:-**
   The MAX30100 is an integrated pulse oximetry and heart-rate monitoring sensor designed by Maxim Integrated.
   It combines two LEDs (Red + IR), a photodetector, optical elements, and analog signal processing in a single compact module.



3. **AD8232 ECG Sensor:-**
   The AD8232 is a single-lead ECG (Electrocardiography) sensor designed by Analog Devices. It is used to measure the electrical activity of the heart



4. **DS18B20 Temperature Sensor**:-
   The DS18B20 is a digital temperature sensor developed by Maxim Integrated.



5. **OLED Display:-**
   An OLED (Organic Light Emitting Diode) Display is a self-emissive screen technology where each pixel emits its own light.



6. **Power Supply (Battery):-**
   A Li-Po (Lithium-Polymer) battery is a lightweight, rechargeable power source commonly used in portable IoT devices, wearables, and health-monitoring systems.
   It provides stable power to components like ESP32, MAX30100, AD8232, DS18B20, and OLED Display.



4.6 SOFTWARE INPUTS

Software / Platform	Usage / Purpose
Arduino IDE	Coding, compiling, and uploading firmware to ESP32
Embedded C	Programming languages used for ESP32
MAX30100, DS18B20, AD8232 Libraries	Sensor interfacing and data acquisition
ThingSpeak	IoT cloud platform for real-time data upload and monitoring

4.7 CODE

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include "MAX30100_PulseOximeter.h"
#include <OneWire.h>
#include <DallasTemperature.h>
#include <WiFi.h>
#include <WiFiClient.h>

// OLED Setup
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET    -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

// MAX30100 Setup
PulseOximeter pox;
#define REPORTING_PERIOD_MS 1000
uint16_t BPM = 0, SpO2 = 0;
uint32_t tsLastReport = 0;

// DS18B20 Setup
#define ONE_WIRE_BUS 23
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);
float tempC = 0.0;

// WiFi & ThingSpeak
const char* ssid = "IOT";
const char* password = "IOT123456";
String apiKey = "UEALCA851CFI756B";
const char* server = "api.thingspeak.com";
WiFiClient client;
unsigned long lastUpdateTime = 0;
const unsigned long updateInterval = 15000; // send every 15s

void onBeatDetected() {
  digitalWrite(2, HIGH);
  delayMicroseconds(500);
  digitalWrite(2, LOW);
}

void setup() {
  Serial.begin(115200);
  pinMode(2, OUTPUT);
```

```
// OLED Init
if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
  Serial.println(F("SSD1306 allocation failed"));
  while(1);
}
display.clearDisplay();
display.setTextSize(1);
display.setTextColor(WHITE);
display.setCursor(0, 0);
display.println("Health Monitoring");
display.println("System Starting...");
display.display();
delay(2000);

// WiFi Init
display.clearDisplay();
display.setCursor(0, 0);
display.println("Connecting WiFi...");
display.display();
WiFi.begin(ssid, password);
while (WiFi.status() != WL_CONNECTED) {
  delay(500);
  Serial.print(".");
}
display.clearDisplay();
display.setCursor(0, 0);
display.println("WiFi Connected!");
display.print("IP: ");
display.println(WiFi.localIP());
display.display();
delay(2000);
display.clearDisplay();

// MAX30100 Init
if (!pox.begin()) {
  Serial.println("MAX30100 INIT FAILED");
  display.clearDisplay();
  display.setCursor(0, 0);
  display.println("MAX30100 Error!");
  display.display();
  while(1);
} else {
  Serial.println("MAX30100 INIT SUCCESS");
  pox.setIRLedCurrent(MAX30100_LED_CURR_7_6MA);
  pox.setOnBeatDetectedCallback(onBeatDetected);
}

// DS18B20 Init
```

```
  sensors.begin();
}

void loop() {
  pox.update();
  unsigned long currentMillis = millis();

  // Update display
  if (currentMillis - tsLastReport > REPORTING_PERIOD_MS) {
    BPM = (pox.getHeartRate() > 0) ? pox.getHeartRate() : BPM;
    SpO2 = (pox.getSpO2() > 0) ? pox.getSpO2() : SpO2;
    sensors.requestTemperatures();
    tempC = sensors.getTempCByIndex(0);

    display.clearDisplay();
    display.setCursor(0, 0);
    display.println("Health Monitor");
    display.printf("HR  : %u bpm\n", BPM);
    display.printf("SpO2: %u %%\n", SpO2);
    display.printf("Temp: %.2f C\n", tempC);
    display.display();

    tsLastReport = currentMillis;
  }

  // Send data to ThingSpeak
  if (currentMillis - lastUpdateTime >= updateInterval) {
    if (WiFi.status() == WL_CONNECTED && client.connect(server, 80)) {
      String postStr = "api_key=" + apiKey;
      postStr += "&field1=" + String(BPM);
      postStr += "&field2=" + String(SpO2);
      postStr += "&field3=" + String(tempC);

      client.println("POST /update HTTP/1.1");
      client.println("Host: api.thingspeak.com");
      client.println("Connection: close");
      client.println("Content-Type: application/x-www-form-urlencoded");
      client.print("Content-Length: ");
      client.println(postStr.length());
      client.println();
      client.println(postStr);
      Serial.println("Data sent to ThingSpeak.");
    } else {
      Serial.println("Failed to connect to ThingSpeak.");
    }
    client.stop();
    lastUpdateTime = currentMillis;
  }
}
```

4.8 ACTUAL COSTING

Component	Function	Approx. Cost (INR)	Notes
ESP32 (NodeMCU ESP32 Dev Board)	Main controller + Wi-Fi connectivity	₹350	Built-in Wi-Fi eliminates need for external module
MAX30100 Sensor	Measures Heart Rate & SpO_2	₹250	Compact and accurate for PPG measurement
AD8232 ECG Module	Captures ECG signals	₹400	Supports 3-lead ECG monitoring
DS18B20 Temperature Sensor	Measures body temperature	₹100	Waterproof and precise
OLED Display (0.96" I^2C)	Displays real-time data	₹150	Low-power and highly readable
Li-ion 18650 Battery + TP4056	Power source + recharge controller	₹200	Rechargeable and portable
Jumper Wires, PCB, Enclosure	Assembly components	₹150	Basic enclosure and wiring

TOTAL COST = 1600

5. ADVANTAGES

Low-Cost and Affordable:
The system uses low-cost sensors and ESP32, making it affordable for rural healthcare centers and economically weaker communities.

Real-Time Remote Monitoring:
Vital health parameters like ECG, heart rate, temperature, and SpO_2 can be monitored remotely in real time through IoT cloud platforms such as ThingSpeak or Blynk.

Reduces Hospital Visits:
Patients do not need to frequently visit hospitals for basic monitoring, reducing travel time, cost, and inconvenience.

Early Detection of Health Issues:
Continuous monitoring enables early detection of abnormal health conditions and timely medical intervention, preventing critical situations.

Useful in Rural and Remote Areas:
Highly suitable for rural and remote regions where hospitals, doctors, and advanced diagnostic facilities are unavailable.

Supports Telemedicine Services:
Enables doctors to monitor patients remotely, offer consultation, and provide treatment recommendations without physical presence.

24×7 Health Tracking:
Offers continuous monitoring of multiple vital signs at any time, unlike traditional manual methods.

Wireless and Mobile Access:
Health data can be accessed from anywhere using smartphone or internet-enabled devices.

Data Storage and Analysis:
Stores patient history on cloud, making it easy to analyze health trends and prepare medical reports.

Energy Efficient and Portable:
Operates on battery or USB power, making the device portable, lightweight, and easy to use in mobile health camps.

Enhances Doctor-Patient Connectivity:
Strengthens communication between healthcare professionals and patients through instant alerts and online consultations.

Scalable and Customizable:
Additional sensors like blood pressure, glucometer, or accelerometer can be added to monitor more health parameters.

Minimizes Human Error:
Digital monitoring reduces manual entry and calculation errors, improving data accuracy.

Supports Emergency Alerts:
Automatically sends alerts via SMS or cloud when readings cross critical thresholds, useful during emergencies.

6. DISADVANTAGES

Internet Dependency:
The system requires stable internet connectivity for real-time monitoring, which may not be available in remote rural areas.

Limited Accuracy Compared to Medical-Grade Devices:
Low-cost sensors may not provide accuracy as high as certified hospital-grade diagnostic devices.

Power Supply Issues in Rural Areas:
Frequent power cuts or lack of battery charging facilities can affect continuous monitoring.

Data Privacy and Security Concerns:
There is a risk of unauthorized access, hacking, or misuse of sensitive patient health data stored on cloud platforms.

Requires Technical Knowledge:
Setup, calibration, and maintenance require basic technical skills that rural users may lack.

Sensor Calibration Required:
Sensors need periodic calibration to maintain accuracy, which might not be easy for non-technical users.

Delay in Emergency Response:
In case of poor internet or slow data transfer, there may be a delay in transmitting critical health alerts to doctors.

Device Maintenance Issues:
Dust, moisture, climatic conditions, or physical damage can affect sensor performance and system reliability.

Limited Local Medical Interpretation:
Even if data is available, lack of trained medical professionals in rural regions may limit proper diagnosis and decision making.

Cloud Service Dependency:
If the IoT cloud service (e.g., ThingSpeak, Blynk) goes down or becomes paid, monitoring may be affected.

Possible False Alarms:
Inaccurate readings due to sensor errors may cause unnecessary panic or misinterpretation.

Not Suitable for Critical Care Patients:
This system is ideal for monitoring stable patients, but may not be reliable for emergency or critical-care scenarios.

7. APPLICATION

1. **Rural Telemedicine Centers**
   - Doctors located in district hospitals can remotely monitor patient vitals (SpO_2, pulse rate, temperature).
   - Useful where doctors are not available physically.

2. Primary Health Centres (PHCs)
   - Health workers (ASHA, ANM) can collect vitals at home and send data to the PHC in real time.
   - Reduces hospital visits for minor monitoring.

3. Home-Based Patient Monitoring
   - For elderly people, chronic patients (heart disease, fever, asthma), who need continuous monitoring.
   - Alerts get sent to family members and doctors.

4. Emergency / Ambulance Monitoring
   - Vitals can be streamed to the hospital while the patient is in transit.
   - Doctors get early information before arrival.

5. COVID-like Pandemic Situations
   - Can monitor oxygen saturation & temperature at isolation centers.
   - Helps detect early deterioration in patients.

6. Maternal and Child Healthcare
   - Monitoring health of pregnant women in remote villages.
   - Temperature, pulse can predict early complications.

7. Industrial & Workplace Health Monitoring
   - Workers in factories, mines, or construction sites can be monitored for overheating or low oxygen.

8. School Health Programs
   - Schools in rural areas can regularly monitor vitals of children at very low cost.

9. Wearable Health Gadgets for Remote Areas
   - Low-cost wearable band for vitals monitoring.
   - Useful in regions with limited access to advanced devices.

10. Smart Village / Digital Health Projects
   - Fits into government initiatives like Digital India, Ayushman Bharat, Telemedicine schemes.

8. FUTURE SCOPE

The proposed IoT-based multi-vital sign monitoring system serves as a foundation for future advancements in smart healthcare and telemedicine. Although the current prototype effectively measures essential parameters such as heart rate, SpO_2, ECG, and body temperature, there is significant potential to expand and enhance its functionality.

In the future, the system can be improved and extended in the following ways:

Integration of Additional Sensors:
Incorporating sensors for blood pressure, glucose levels, respiratory rate, and body motion can provide a more comprehensive health profile, allowing continuous monitoring of patients with chronic conditions.

Artificial Intelligence (AI) and Machine Learning (ML):
Implementing AI/ML algorithms will enable predictive health analytics and early disease detection by identifying abnormal patterns or trends in physiological data over time.

Mobile Application Development:
A dedicated Android or iOS mobile application can be developed to provide patients and doctors with a user-friendly interface for real-time data visualization, health alerts, and personalized health reports.

Cloud Data Security and Integration:
Future versions can focus on secure data transmission and storage using advanced encryption techniques to ensure patient privacy and data integrity, in compliance with healthcare data standards.

Wearable Device Implementation:
The current prototype can be transformed into a compact wearable device (e.g., wristband or patch) for continuous, real-time health monitoring without restricting patient mobility.

Integration with Telemedicine Platforms:
The device can be linked with existing telemedicine systems to enable live consultations, remote prescriptions, and emergency alerts, thereby strengthening digital healthcare infrastructure.

By incorporating these advancements, the system can evolve into a complete IoT-based smart healthcare solution that provides predictive, preventive, and personalized medical support. This will not only benefit rural and underserved areas but also contribute to the digital transformation of global healthcare systems.

9. ACTUAL CIRCUIT

10.CONCLUSION

The developed IoT-based multi-vital sign monitoring system effectively measures key health parameters such as heart rate, SpO_2, ECG, and temperature, enabling real-time remote monitoring through the ThingSpeak cloud. This low-cost, portable device enhances healthcare accessibility, especially in rural areas, by bridging the gap between patients and doctors. Overall, the project demonstrates how IoT technology can revolutionize healthcare through continuous monitoring, timely intervention, and improved patient care.

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